BIOCHEMISTRY CUME
Environmental Effects on Biomacromolecules
April 15, 2011

1. Examine the following peptide sequence:

Leu-Lys-Arg-Phe-Asp-His-Tyr-Glu-Arg-Ala-Gly-Pro-Glu-His-Lys-Ile

At pH = 6.0, estimate the overall net charge of this peptide. Explain in detail the rationale for your answer (25 pts).

2. The titration curve of alanine shows the ionization of two functional groups with pKa values of 2.34 and 9.69. The titration curves of di-, tri-, and longer oligopeptides of alanine also show the ionization of only two functional groups, although the experimental pKa values are different.

Amino Acid or Peptide	pK1	pK2
Ala	2.34	9.69
Ala-Ala	3.12	8.30
Ala-Ala-Ala	3.39	8.03
Ala-$(Ala)_n$-Ala, n>4	3.42	7.94

A) Draw the structure of Ala-Ala-Ala, showing all atoms. Identify functional groups associated with pK1 and pK2 (20 pts).

B) Explain why the value of pK1 increases with addition of Ala residues in the Ala oligopeptide sequence (20 pts).

C) Provide a rationale for the finding that the value of pK2 decreases with each addition of an Ala residue to the Ala oligopeptide sequence. (20 pts).

3. At the pI (isoelectric point) of a biomacromolecule, such as a protein, one often finds that the macromolecule displays its lowest level of solubility. Explain this finding (15 pts) in as much detail as time and knowledge permit.